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The Parathyroid Glands

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Bailey & Love Bailey & Love Bailey & Love PART 8 | Endocrine and breast

CHAPTER

56 The parathyroid glands

Learning objectives

To understand:

- The anatomy of the parathyroid glands
- The physiology of calcium regulation
- The underlying causes of hypercalcaemia and appropriate emergency management
- The aetiology, presentation, investigation and management of primary hyperparathyroidism and associated special cases
- The aetiology, presentation, investigation and management of secondary and tertiary hyperparathyroidism
- The aetiology and management of parathyroid carcinoma

INTRODUCTION

The parathyroid glands were first described by Sir Richard Owen in a neck dissection of an Indian rhinoceros at the London Zoological Gardens in 1850. Credit for recognition of the 'glandulae parathyreoidae' goes, however, to Sandström, who published a monograph in 1887 on dissection of the parathyroid glands and their blood supply in animals and human cadavers. Unfortunately, Sandström committed suicide at the age of 37 and it was not until the 1890s that his work was rediscovered by Gley, who associated tetany following thyroid surgery with removal of the parathyroid glands. In 1905, MacCallum found that he could relieve postoperative tetany by the injection of parathyroid extract. While the association between parathyroid enlargement and bone disease was reported in 1907, it was not until 1925 that the first parathyroidectomy was performed by Mandl in Vienna on Albert Gahne, a tram conductor with severe primary hyperparathyroidism (PHPT) and osteitis fibrosa cystica.

ANATOMY OF THE PARATHYROID GLANDS

The developmental embryology and surgical anatomy of the parathyroid glands are intimately linked, and knowledge of both is essential for successful surgical treatment of parathyroid disease.

The parathyroid glands, of which there are four, develop from the third and fourth pharyngeal pouches between the

fifth and 12th weeks of gestation. They are typically described as 'Portland brick' (yellow/brown) in colour and weigh approximately 30 mg. Approximately 13% of the population have abnormal parathyroid tissue, with 5% having a true supernumerary gland. The blood supply of both the superior and inferior parathyroid glands arises from the inferior thyroidal artery. While the location of the individual glands may vary significantly, there appears to be a degree of symmetry between opposite sides that can be helpful during surgical dissection.

The inferior parathyroid gland and the thymus arise from the third pharyngeal pouch. As a result of the longer normal embryological descent, there is correspondingly more variation in their anatomical position. However, in more than 50% of cases they are located at the inferior pole of the thyroid gland, on the anterior, lateral or posterior surface. The gland itself is freely mobile within a globule of fat adjacent to the lower pole (*Figure 56.1a*).

The superior parathyroid glands arise from the dorsal portion of the fourth pharyngeal pouch. As a result of their more limited embryological descent they are more constant in position. In more than 80% of patients, the superior parathyroid glands are located at the posterior aspect of the thyroid lobe in an area 2 cm in diameter, centred 1 cm around the junction of the inferior thyroid artery and the recurrent laryngeal nerve in strict proximity to the cricothyroid junction (*Figure 56.1b*). The parathyroid glands are closely associated with, but contained within, a halo of fat that is freely mobile over the thyroid capsule.

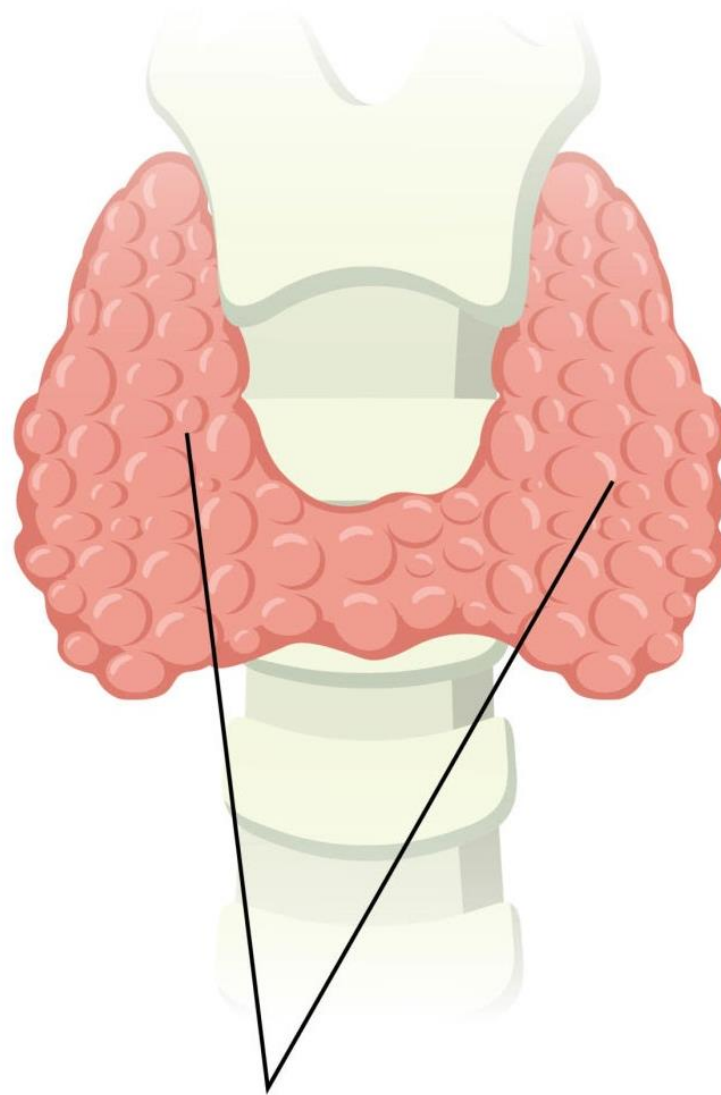
Sir Richard Owen, 1804–1892, English comparative anatomist and palaeontologist. First director of the Natural History Museum, London, UK, and Hunterian Professor at the Royal College of Surgeons of England.

Ivar Viktor Sandström, 1852–1889, medical student, Uppsala, Sweden.

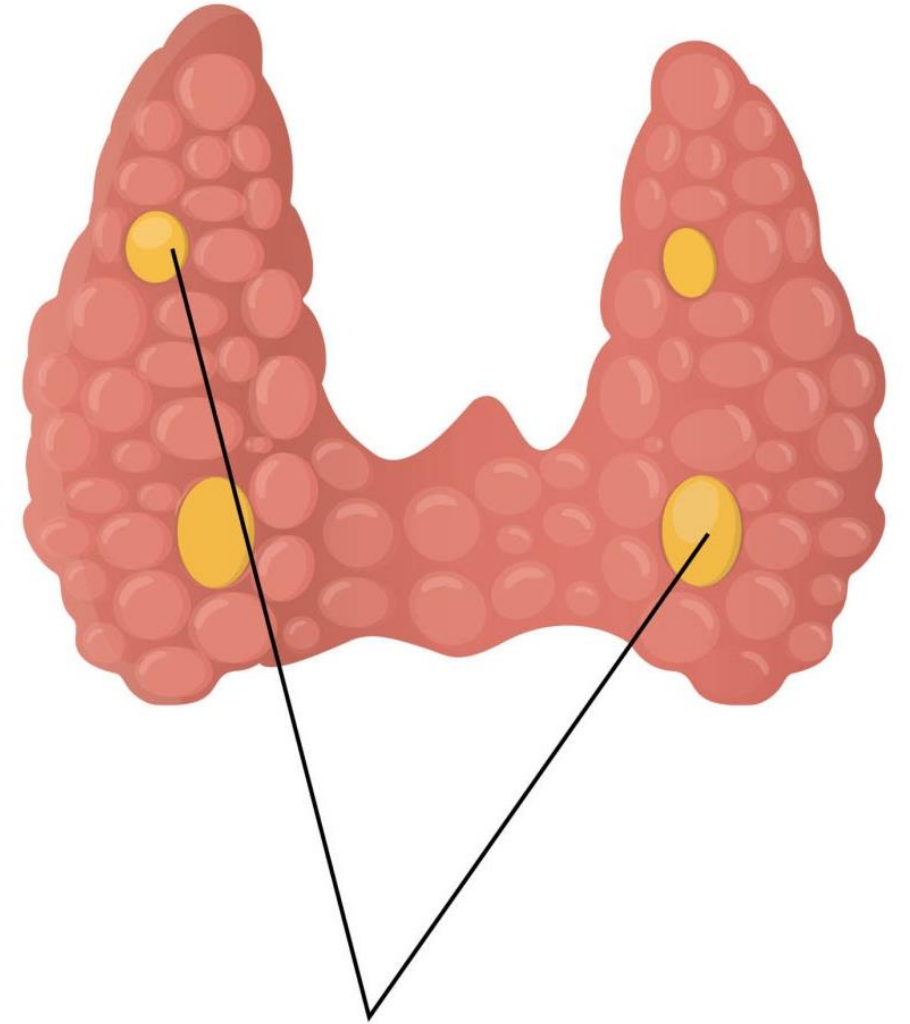
Marcel Eugene Gley, 1857–1930, French pathologist.

William J MacCallum, 1874–1944, Professor of Pathology, Johns Hopkins Hospital, Baltimore, MD, USA.

Felix Mandl, 1892–1957, Professor of Surgery, Vienna, Austria.



Thyroid gland



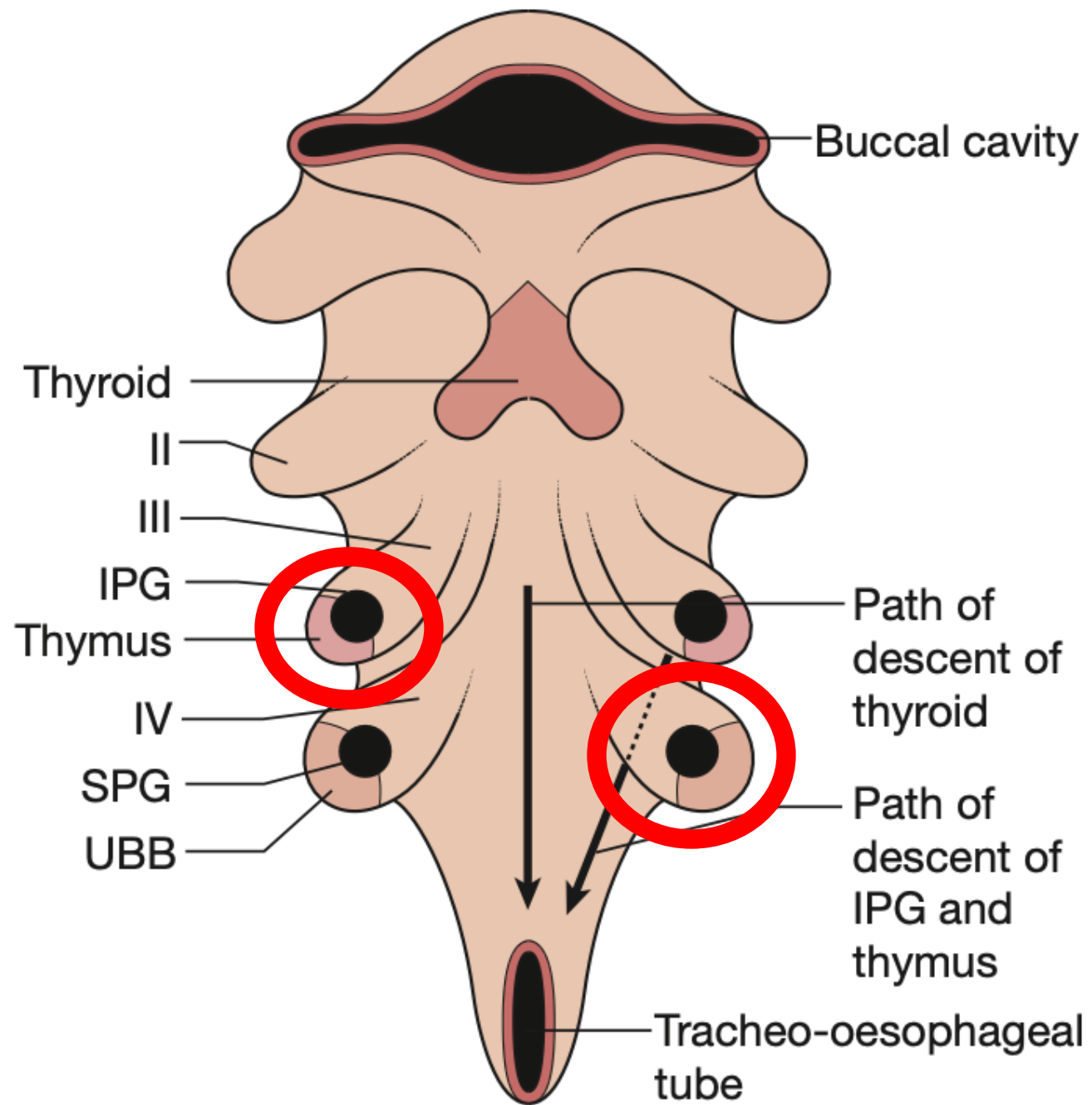
Parathyroid glands

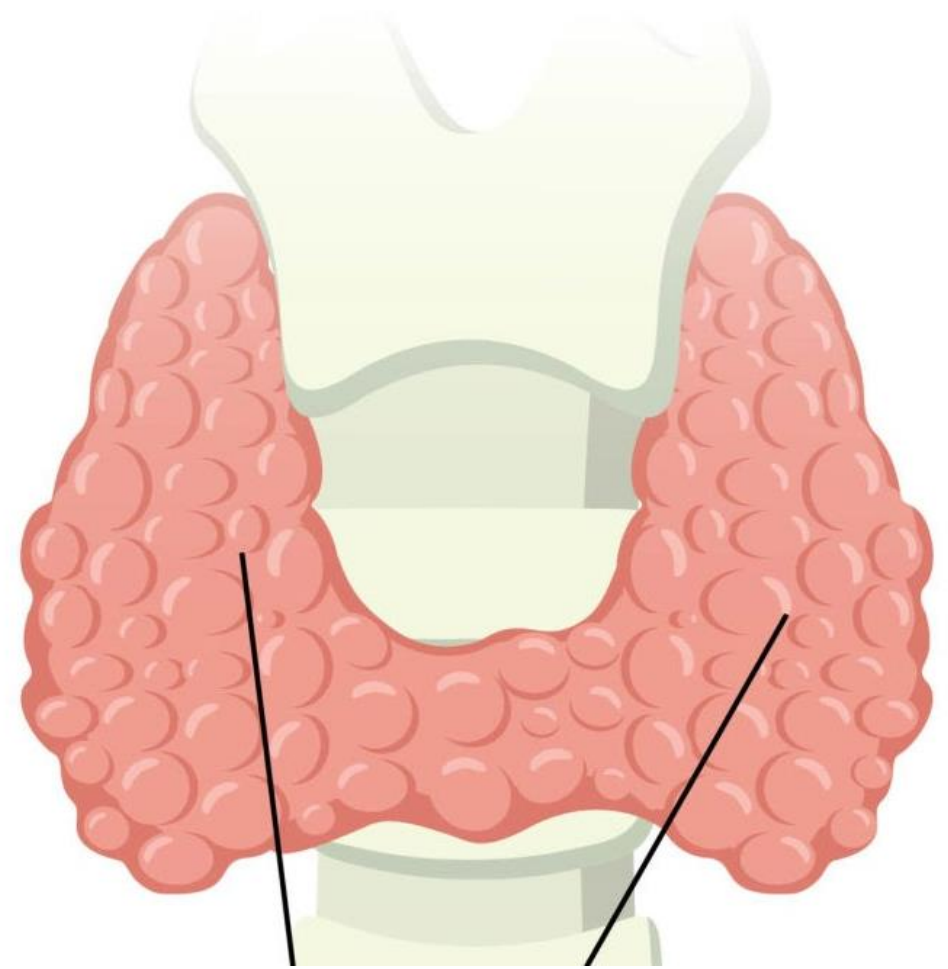
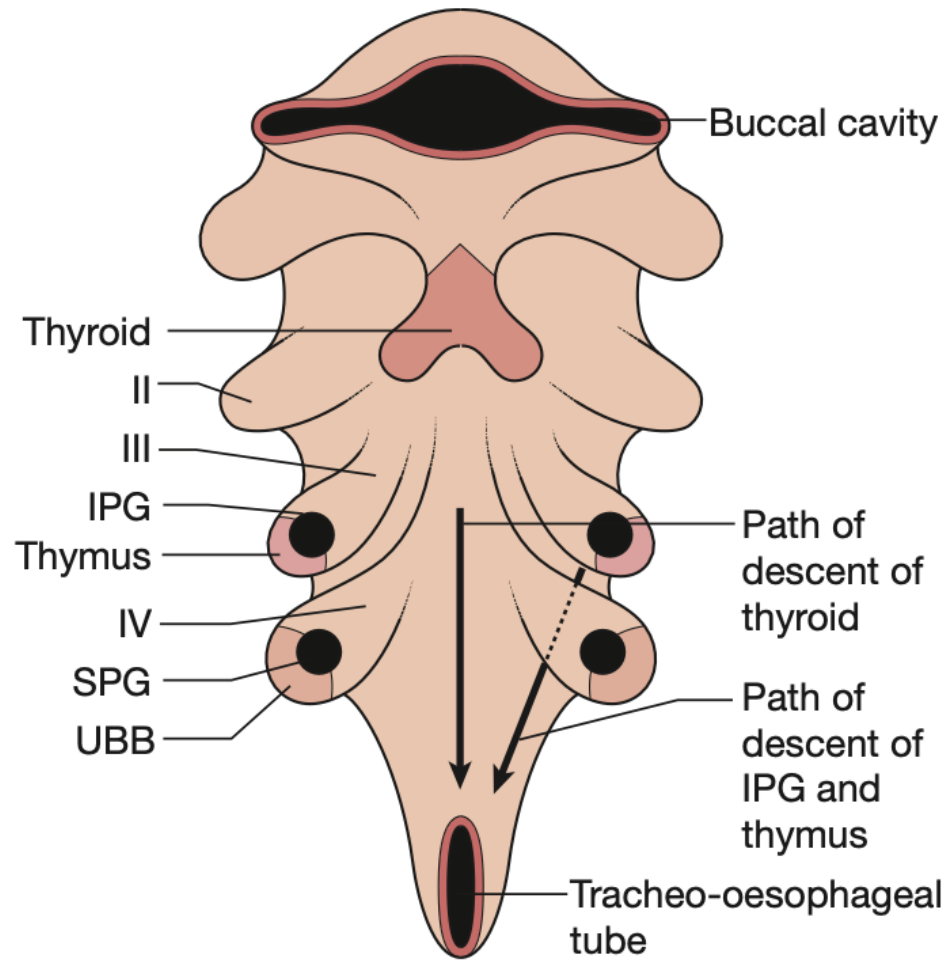


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Anatomy of the Parathyroid Glands

- The parathyroid glands develop from the third and fourth pharyngeal pouches







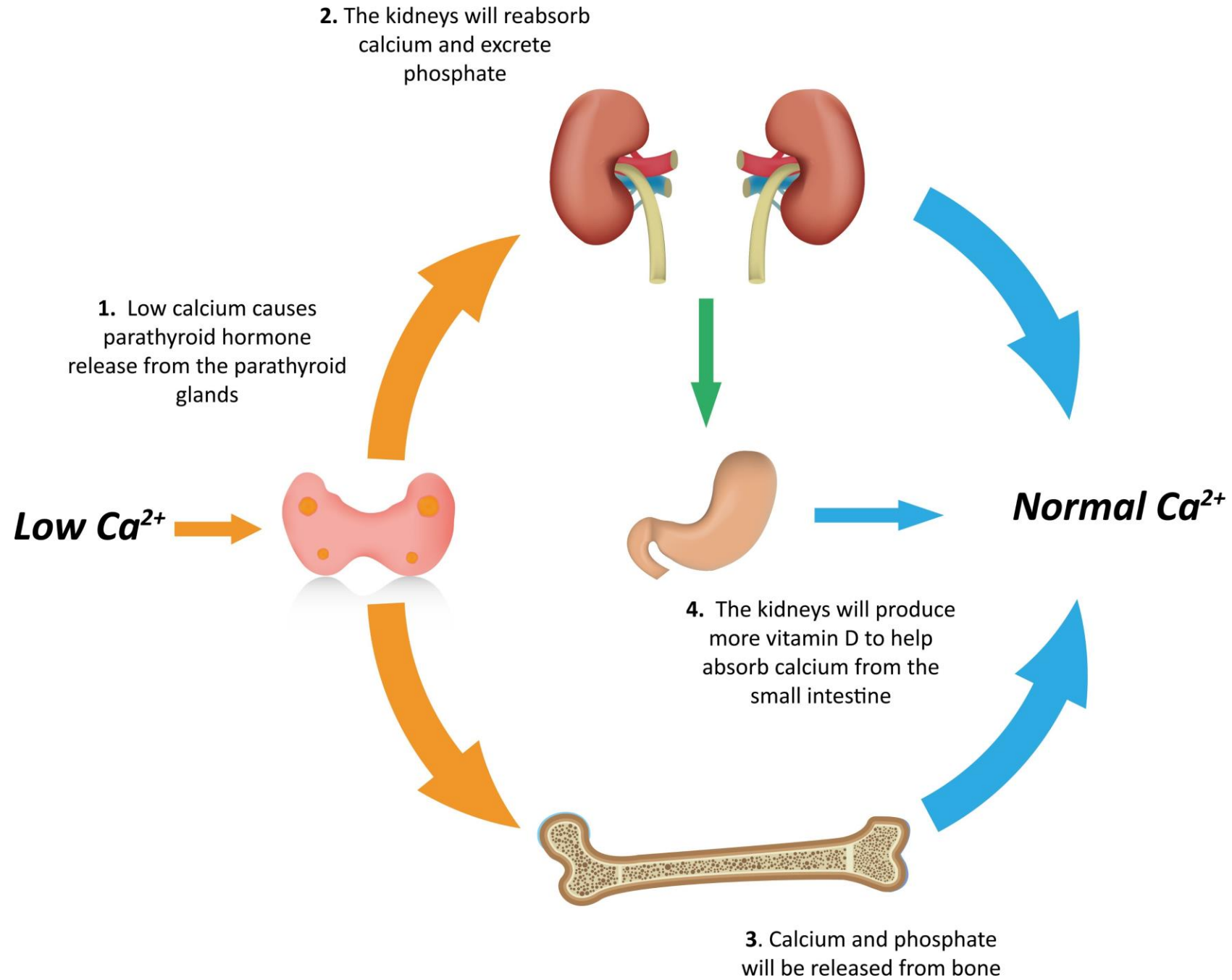
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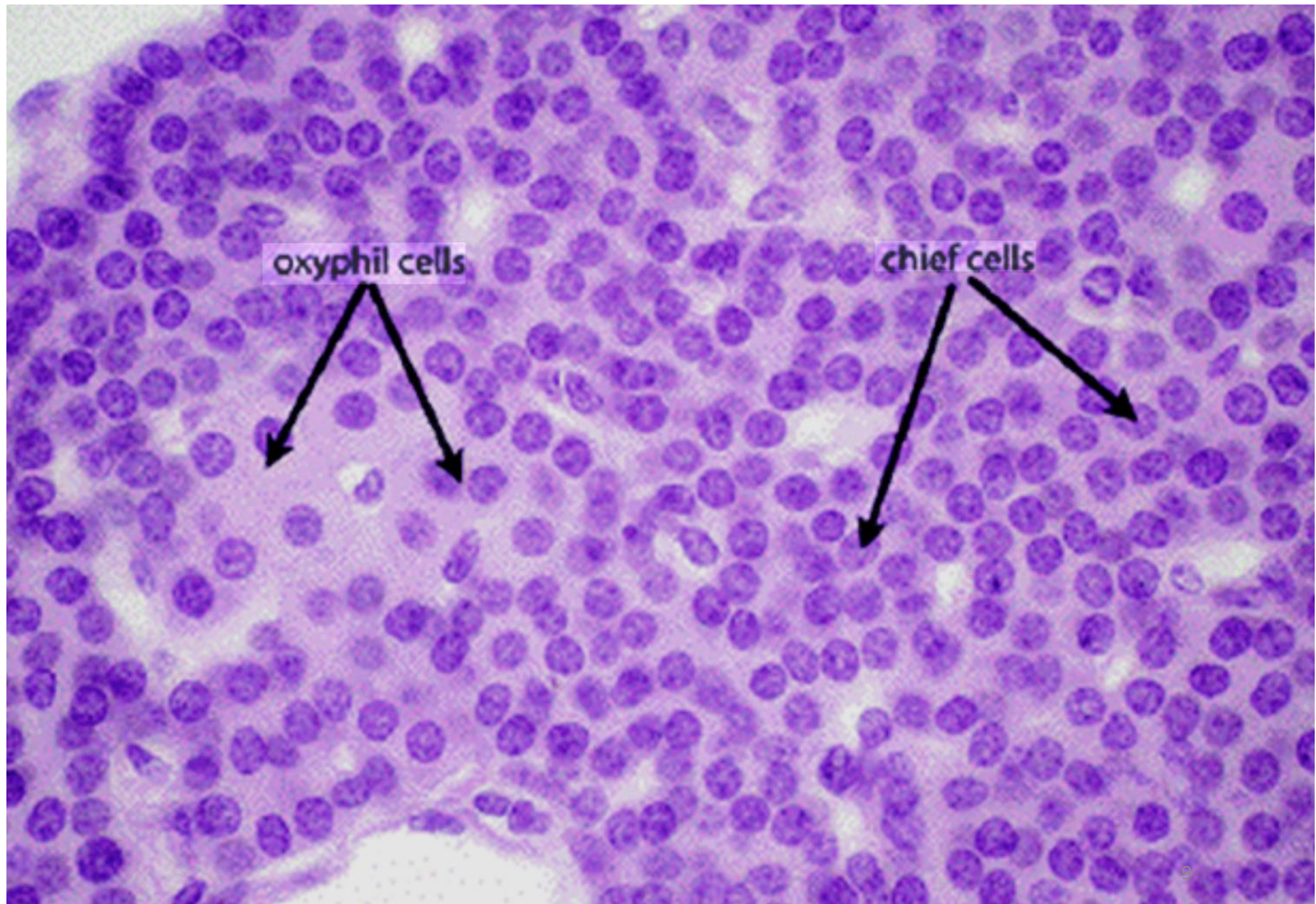
Parathyroid Hormone (PTH) function

PTH's main function is calcium regulation, and it indirectly affects magnesium.

Low serum calcium or high serum phosphate and magnesium stimulate PTH secretion.

PTH has a half-life of 3-5 min.







- Chief cells are sensitive to low ionized calcium and high phosphate levels in the blood
- This increases the PTH production



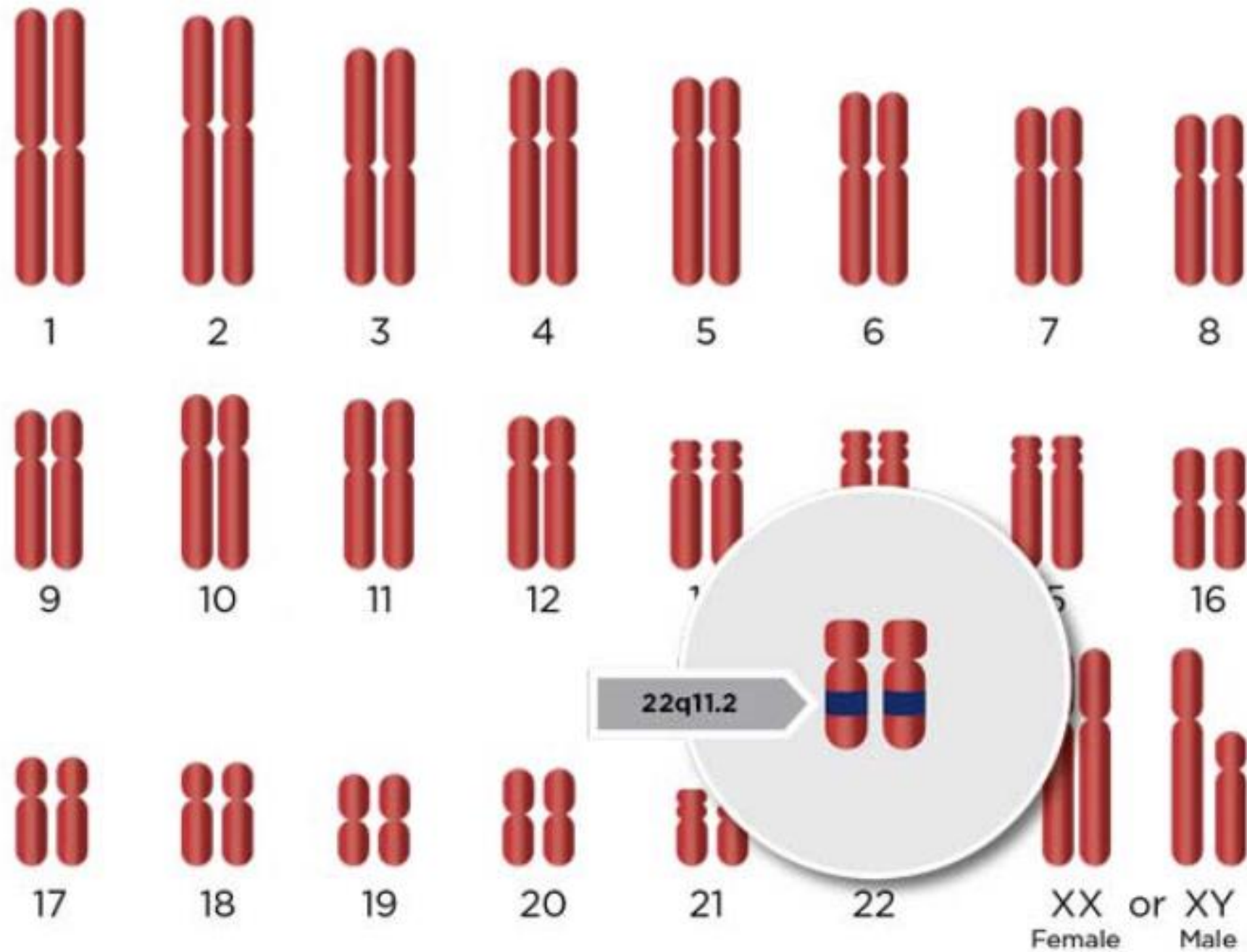
Causes of hypoparathyroidism

1. Surgical removal during thyroidectomy and parathyroidectomy
2. Autoimmune causes

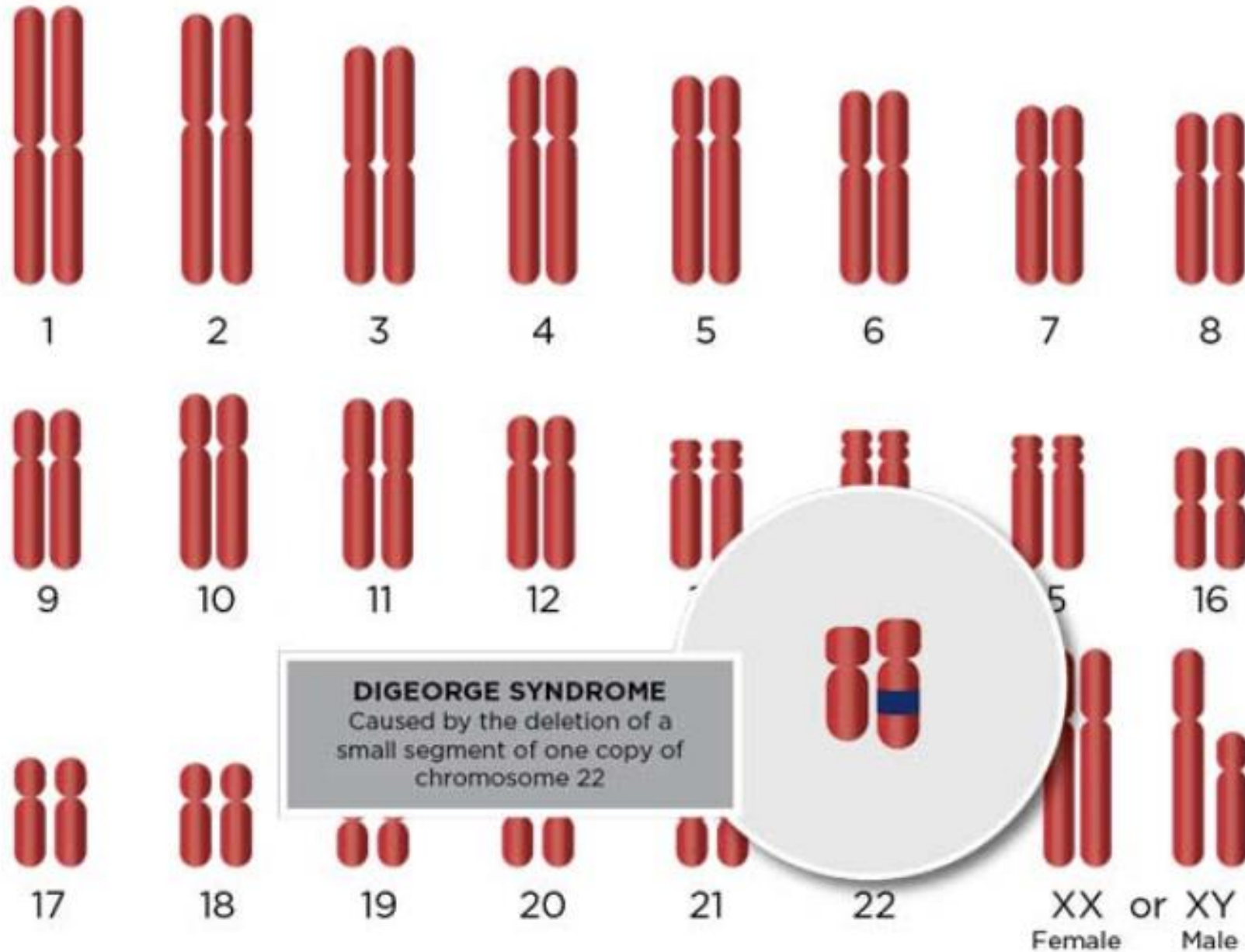


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3. Genetic causes (DiGeorge syndrome, failure of 3rd and 4th pharyngeal pouch development)



DIGEORGE CHROMOSOMES





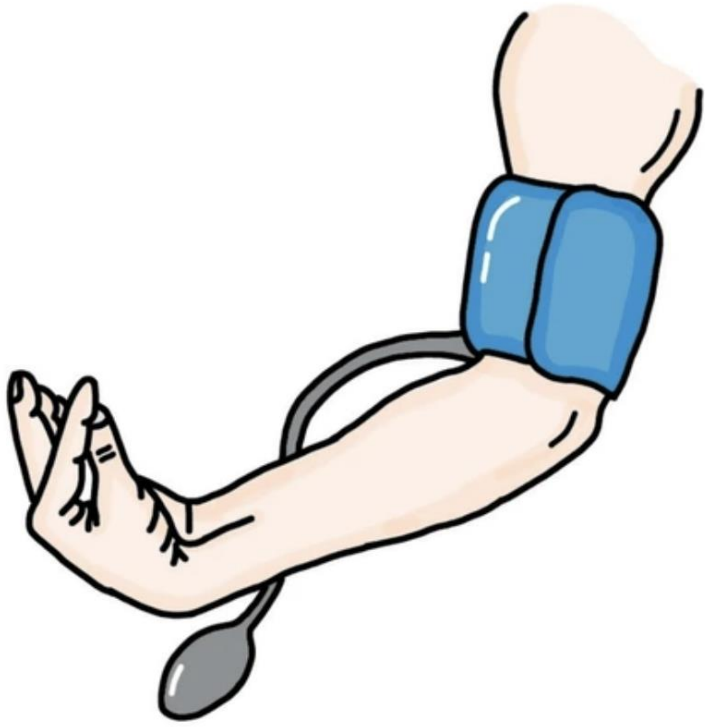
4. Pseudo-hypothyroidism (normal or high PTH), kidneys, and bones don't respond to PTH (Albright osteodystrophy)



Clinical features

- Hypocalcemia and hypophosphatemia
 1. Involuntary muscle contraction.
 2. Tetany.
 3. Chvostek's sign.
 4. Trousseau sign.





Trousseau's sign

Trousseau's sign
Armand Trousseau, French internist



- **Chvostek's sign:-** is the involuntary twitching of facial muscles when the facial nerve is tapped, indicating neuromuscular irritability.



František Chvostek

Czech-Austrian military physician and lecturer in internal medicine 1876





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- PTH ↓
- Ca^{+2} ↓
- Phosphate (PO_4^{-3}) ↑



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Hyperparathyroidism



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1. Bones.
2. Stones.
3. Thrones.
4. Abdominal groans.
5. Psychiatric overtones.



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- Bone pain
- Muscle weakness
- Constipation
- Renal stone (Calcium stone)
- Depression



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Types

1. Primary Hyperparathyroidism (PHPT)
2. Secondary Hyperparathyroidism (SHPT)
3. Tertiary Hyperparathyroidism (THPT)



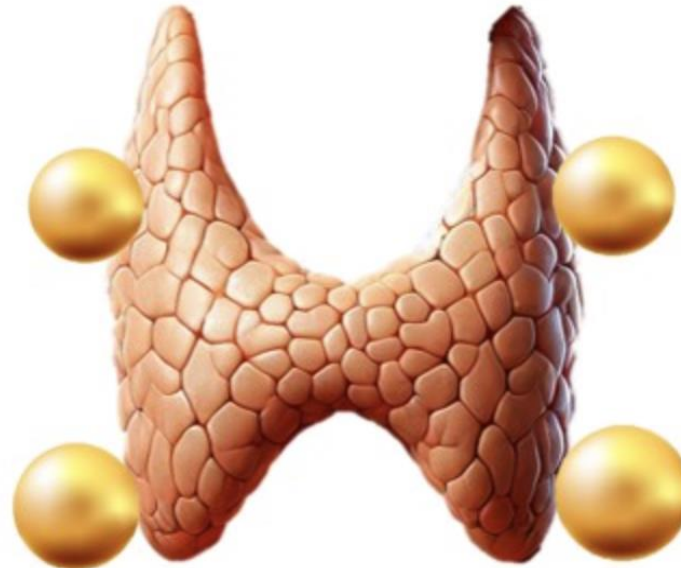
Normal parathyroid glands



Single enlarged parathyroid gland (adenoma)



2 enlarged parathyroid glands
("double adenoma")



Parathyroid hyperplasia (all 4
glands enlarged)



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- Primary hyperthyroidism
- Adenoma is the commonest > 80%



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Multiple Endocrine Neoplasia (MEN)

- MEN 1 (PPP)

MEN 2a (PPM)

MEN 2b (PPM)



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MEN 1 PPP

pituitary hyperplasia
pancreatic tumor gastrinoma
parathyroid adenoma.



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MEN 2A (PPM)

parathyroid hyperplasia.
Pheochromocytoma
Medullary thyroid tumor



- MEN 2b (PMM)
- Pheochromocytoma
- Medullary thyroid tumor
- Mucosal neuromas (benign nerve tumors) on the lips, tongue, and eyelids
- Ganglioneuromas in the gastrointestinal tract



Secondary hyperparathyroidism

- Chronic kidney disease
- Low vit D

- PTH



- Ca^{+2}



- Phosphate (PO_4^{-3})





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Blood test

- Ca^{+2} ↑
- Phosphate (PO_4^{-3}) ↓
- PTH ↑



Tertiary hyperparathyroidism

- Only in end-stage kidney disease.
- Hyperplasia of the parathyroid gland that are autonomous



Blood test

- Ca^{+2} ↑
- Phosphate (PO_4^{-3}) ↑
- PTH ↑



Osteitis fibrosa cystica

- Brown tumors of the long bones and associated subperiosteal bone reabsorption.
- Distal tapering of the clavicles.
- Salt and pepper erosions of the skull.





- Renal stones in over 80% of
- Neuromuscular dysfunction and muscle weakness.



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Localization Studies

Localization is Crucial for minimally invasive surgery.

Sestamibi Scan (Nuclear Medicine):
Most accurate/reliable.

Ultrasonography (US): Non-invasive, inexpensive.
Operator-dependent.



4D-CT Scan: Provides anatomical and functional information (rapid uptake/washout helps differentiate from lymph nodes).

Invasive (Reserved for Reoperative Surgery):
Fine-needle aspiration with PTH assay, selective venous sampling for PTH gradient.



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